

Final Technical Audit

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Audit date: July 7, 2026

Deployed demo: <https://crossroad-fhir-link-three.vercel.app>

Automated Click-Test Result

Status: PASS

Routes tested:

- USA -> India
- India -> USA
- Australia -> Europe
- Europe -> USA

For each route, the audit verified:

- selected source and target state match the UI
- Clinical Trace renders
- FHIR IPS Bundle tab renders
- Evidence & Limitations tab renders
- source PDF downloads as a valid PDF
- final target-country PDF downloads as a valid PDF
- FHIR Bundle JSON downloads and parses
- evidence pack JSON downloads and parses
- FHIR Bundle JSON has `resourceType=Bundle`
- FHIR Bundle JSON has `type=document`
- first Bundle entry is `Composition`

Audit file: click-test-audit.json

UI Status

No broken route was detected in the automated browser audit.

Screenshots were captured for:

- each route dashboard
- each route FHIR Bundle view
- each route evidence view
- mobile dashboard smoke test

Claim Check

Approved wording:

- "FHIR IPS is the interoperable artifact; PDFs are human-readable renderings."
- "Readiness checks only; no national profile certification claimed."
- "FedAvg gives data locality only; model updates can leak information without DP-SGD or secure aggregation."

Avoided wording:

- "certified national conversion"
- "fully private federated learning"
- "production-ready clinical decision support"

Evidence Consistency

Metrics shown in the UI match the downloadable evidence pack:

- 20 synthetic reports
- 4 sites
- 768 terminology training examples
- 48 cross-site transfer examples
- 192 globally unseen examples
- semantic transfer validation: 48/48 correct versus 47/48 local-only (+1)
- globally unseen benchmark: 192/192 correct, macro-F1 1.000; local-only average accuracy 94.4%
- official HL7 FHIR Validator 6.9.11 against IPS 2.0.1: 4/4 Bundles, 0 errors, 0 warnings
- informational notes: 2 per Bundle because RxNorm ingredients are outside the IPS guide's recommended medication value set

The model metrics are deterministic single-seed synthetic results and are not clinical performance evidence.

Remaining Submission Risks

These should be stated openly to judges:

- data is synthetic
- extraction is rule-backed
- terminology operations are simulated
- national checks are readiness-only
- FL does not provide formal privacy without additional safeguards

Final Freeze Recommendation

Freeze implementation now unless a critical bug appears.

Focus remaining time on:

- demo video

- narrative clarity
- final submission form
- Q&A; preparation